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Thistle funnel experiment

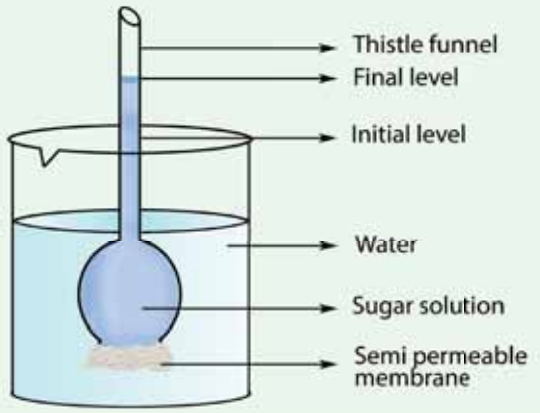


Figure 11.6: Thistle Funnel Experiment

Mouth of a thistle funnel is tied with goat bladder. It acts as a semipermeable membrane. Pour concentrated sugar solution in the thistle funnel and mark the level of solution. Place this in a beaker of water. After some time, water level in the funnel rises up steadily. This is due to the inward diffusion of water molecules through the semipermeable membrane (Figure 11.6).

Conversely, if water in the beaker is replaced by a sugar solution and sugar solution in the thistle funnel replaced by water, what will be happen?

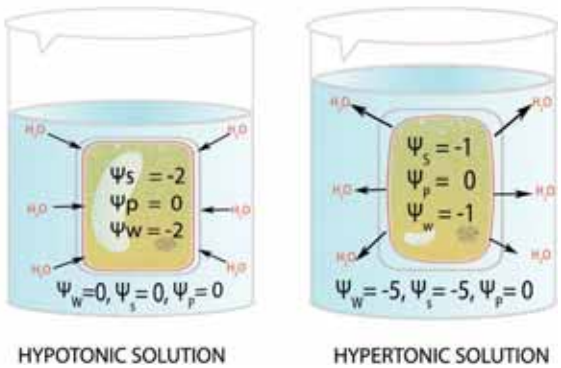


Figure 11.7: Types of solution based on concentration

in comparison with another solution.

1. Types of osmosis

Based on the direction of movement of water or solvent in an osmotic system, two types of osmosis can occur, they are ***Endosmosis*** and ***Exosmosis***.

i. ***Endosmosis***: Endosmosis is defined as the osmotic entry of solvent into a cell or a system when it is placed in a pure water or hypotonic solution.

For example, dry raisins (high solute and low solvent) placed in the water, it swells up due to turgidity.

ii. ***Exosmosis***: Exosmosis is defined as the osmotic withdrawal of water from a cell or system when it is placed in a hypertonic solution. Exosmosis in a plant cell leads to **plasmolysis**.

2. Plasmolysis (*Plasma* = cytoplasm; *lysis* = breakdown)

When a plant cell is kept in a hypertonic solution, water leaves the cell due to **exosmosis**. As a result of water loss, protoplasm shrinks and the cell membrane is pulled away from the cell wall and finally, the cell becomes **flaccid**. This process is named as **plasmolysis**.

Wilting of plants noticed under the condition of water scarcity is an indication of plasmolysis. Three types of plasmolysis occur in plants: i) ***Incipient plasmolysis ii) Evident plasmolysis*** **and** ***iii) Final plasmolysis***. Differences among them are given in table 11.2.

Significance

Plasmolysis is exhibited only by living cells and so it is used to test whether the cell is living or dead.